game44

a

capacitor acts as a open at t=0-
inductor acts as a short circuit

$$i_L(0^-) = 20 = \boxed{20mA}$$

b

Current through an inductor cannot change instantaneously

$$i_L(0^+) = i_L(0^-) = \boxed{20mA}$$

c

Combine 3V and 150 ohm to norton equivalent

$$I_N = 20mA$$

Inductor acts as a short, capacitor acts as an open

$$i_L(\infty) + 120mA - 20mA = 0$$

$$i_L(\infty) = \boxed{-100mA}$$

d

$\frac{d}{dt} i_L(0^+) = \frac{V_0}{L}$ for all cases, over, under, or critically damped.

voltage across parallel components is the same, so $V_L(0^+) = V_C(0^+) = 0V$

$$\frac{V_0}{L} = \frac{0}{31.25mH} = \boxed{0\frac{A}{s}}$$

e

$$R_{eq} = 300||150 = 100\Omega$$
$$\alpha = \frac{1}{2RC} = \boxed{10,000 s^{-1}}$$
$$\omega = \sqrt{\frac{1}{LC}},\ \omega = 8,000 s^{-1}$$
$\alpha^2 > \omega^2$, so the circuit is overdamped.
$$\sqrt{\alpha^2 - \omega_0^2} = 6,000 \frac{rad}{s}$$
$$s_1 = -\alpha + 6000 = -4,000 \frac{rad}{s}$$
$$s_2 = -\alpha - 6000 = -16,000 \frac{rad}{s}$$
$$i_L(0^+) = I_f + A_1' + A_2'$$
$$A_1' + A_2' = 120mA$$
$$0 = s_1 A_1' + s_2 A_2'$$
$$0 = -4000 A_1 - 16000 A_2$$
$$A_1 = -4A_2$$
$$-3A_2 = 120mA$$
$$A_2 = -40mA$$
$$A_1 = 160mA$$
$$i_L(t) = \boxed{-100 + 160e^{-4000t} - 40e^{-16000t} mA}$$